

LAB 0

Aim: To study SQL commands, their uses, syntax, and install MySQL for database management experiments.

1. What is SQL and What are its Uses?

SQL (Structured Query Language) is a domain-specific language used to store, retrieve, manipulate, and manage data in a Relational Database Management System (RDBMS).

SQL is mainly used for handling structured data, where data is organized into tables consisting of rows and columns. SQL is generally a non-case-sensitive language.

Uses of SQL:

- SQL can retrieve data from a database.
- SQL can insert new data into a database.
- SQL can create new databases.
- SQL can create new tables in databases.
- SQL can update existing data in databases.
- SQL can delete data from databases.
- SQL can perform queries related to databases.
- SQL can create and manage database objects such as tables, views, and indexes.
- SQL can set permissions on tables, procedures, views, and other database objects.

SQL Standards:

SQL is an American National Standards Institute (ANSI) and International Organization for Standardization (ISO) standard.

Although SQL is an ANSI/ISO standard, different database management systems may have different versions and extensions of SQL. However, most of them support the major SQL commands, such as SELECT, UPDATE, DELETE, INSERT, and WHERE, in a similar manner.

2. Different Commands of SQL, Their Uses and Syntax

The following are some of the commonly used SQL commands:

1. SELECT Command Use:

The SELECT command is used to retrieve data from a database table.

Syntax:

```
SELECT column1, column2, ... FROM table_name;
```

To select all columns:

```
SELECT *FROM table_name;
```

2. UPDATE Command

Use:

The UPDATE command is used to modify existing data in a database table. Syntax:

```
UPDATE table_name  
SET column1 = value1, column2 = value2, ... WHERE condition;
```

3. DELETE Command Use:

The DELETE command is used to delete existing records from a database table. Syntax:

```
DELETE FROM table_name WHERE condition;
```

4. INSERT INTO Command Use:

The INSERT INTO command is used to insert new records into a database table.

Syntax:

```
INSERT INTO table_name (column1, column2, column3, ...) VALUES (value1, value2, value3, ...);
```

5. CREATE DATABASE Command

Use:

The CREATE DATABASE command is used to create a new database.

Syntax:

```
CREATE DATABASE [IF NOT EXISTS] database_name [CHARACTER SET charset_name]  
[COLLATE collation_name];
```

6. ALTER DATABASE Command

Use:

The ALTER DATABASE command is used to modify the properties or settings of an existing database.

Syntax:

```
ALTER DATABASE [database_name] [DEFAULT] CHARACTER SET charset_name  
[DEFAULT] COLLATE collation_name;
```

7. CREATE TABLE Command Use:

The CREATE TABLE command is used to create a new table in a database.

Syntax:

```
CREATE TABLE table_name (  
column1_name data_type(size) [constraints],  
column2_name data_type(size) [constraints],  
...  
[table_constraints]  
);
```

8. ALTER TABLE Command Use:

The ALTER TABLE command is used to modify the structure of an existing table. It can be used to add, modify, or remove columns and constraints.

Syntax:

```
ALTER TABLE table_name action1,  
action2,  
...;
```

9. DROP TABLE Command

10.

Use:

The DROP TABLE command is used to permanently delete a table and all its data from a database.

Syntax:

```
DROP TABLE [IF EXISTS] table_name;
```

11. CREATE INDEX Command

Use:

The CREATE INDEX command is used to create an index on one or more columns. An index can improve the speed of data retrieval operations.

Syntax:

```
CREATE [UNIQUE | FULLTEXT | SPATIAL] INDEX index_name  
ON table_name (column1 [ASC | DESC], column2 [ASC | DESC], ...);
```

12. DROP INDEX Command Use:

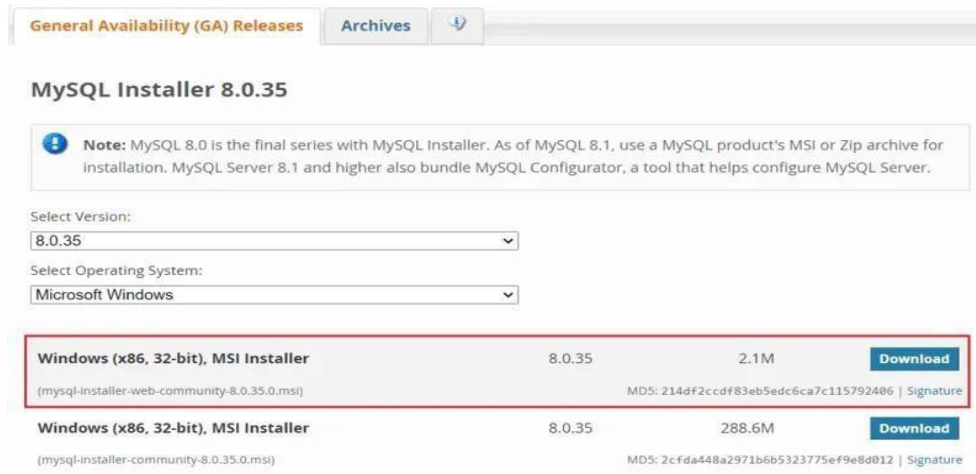
The DROP INDEX command is used to remove an existing index from a table. Syntax:

DROP INDEX index_name ON table_name;

MySQL Installation:

Step 1 – Download MySQL Installer

Select the Windows MSI Installer download.



The screenshot shows the MySQL Community Downloads page for the MySQL Installer 8.0.35. It includes a note about MySQL 8.0 being the final series with the installer, and a table of download options for Windows (x86, 32-bit) MSI installers. The first option is highlighted with a red box.

Operating System	Version	Size	Action
Windows (x86, 32-bit), MSI Installer (mysql-installer-web-community-8.0.35.0.msi)	8.0.35	2.1M	Download
Windows (x86, 32-bit), MSI Installer (mysql-installer-community-8.0.35.0.msi)	8.0.35	288.6M	Download

Step 2 – Start the Download

Select “No thanks, just start my download.”



The screenshot shows the MySQL Community Downloads page. It includes a login/sign-up section and a link to start the download without logging in. The link is highlighted with a red box.

[No thanks, just start my download.](#)

Step 3 – Wait for the Installer

Wait while Windows configures MySQL Installer.



The screenshot shows the MySQL Installer - Community window. It displays a progress bar and a 'Cancel' button. The text 'Please wait while Windows configures MySQL Installer - Community' is visible.

MySQL Installer - Community

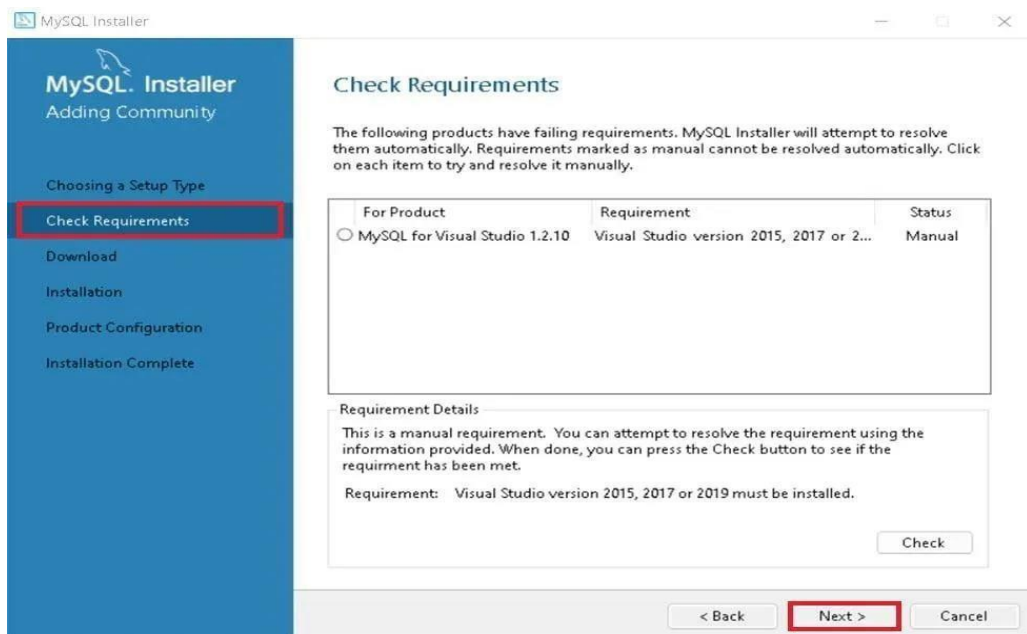
Please wait while Windows configures MySQL Installer - Community

Time remaining: 14 minutes

Cancel

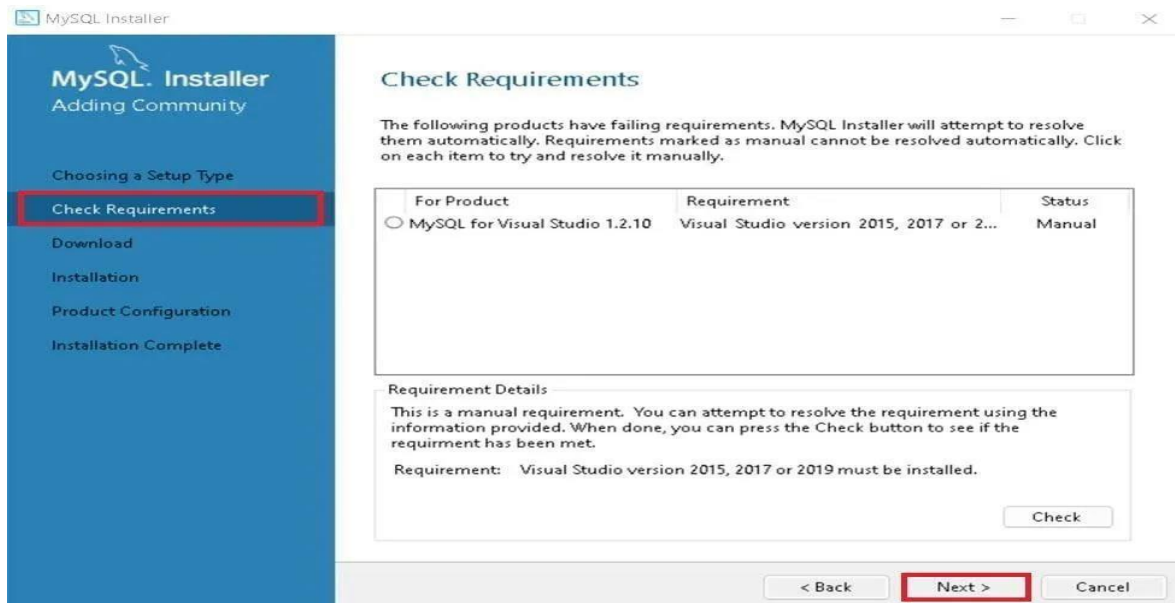
Step 4 – Check Requirements

Check the listed requirements and continue with Next.



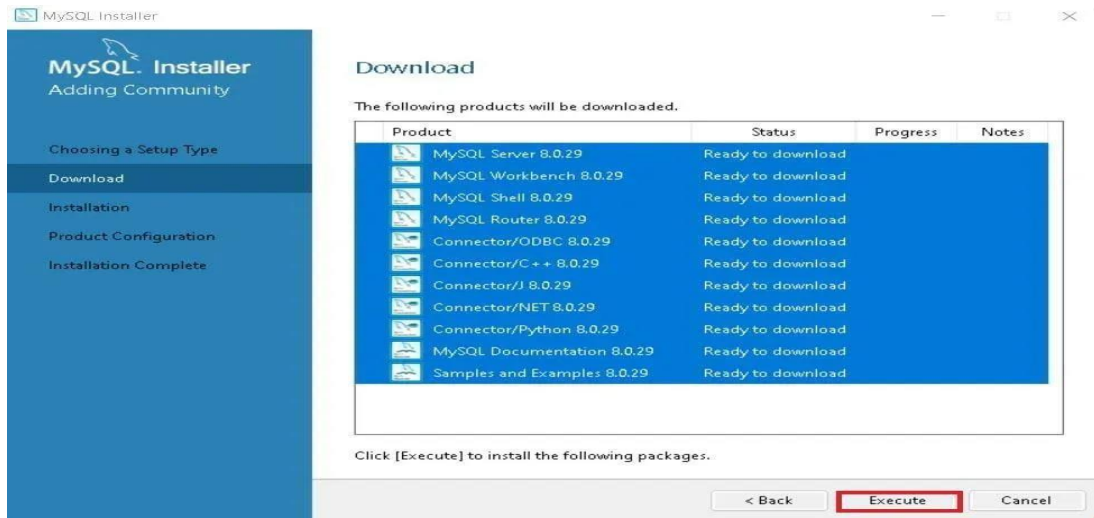
Step 5 – Continue

Continue to the Download section.



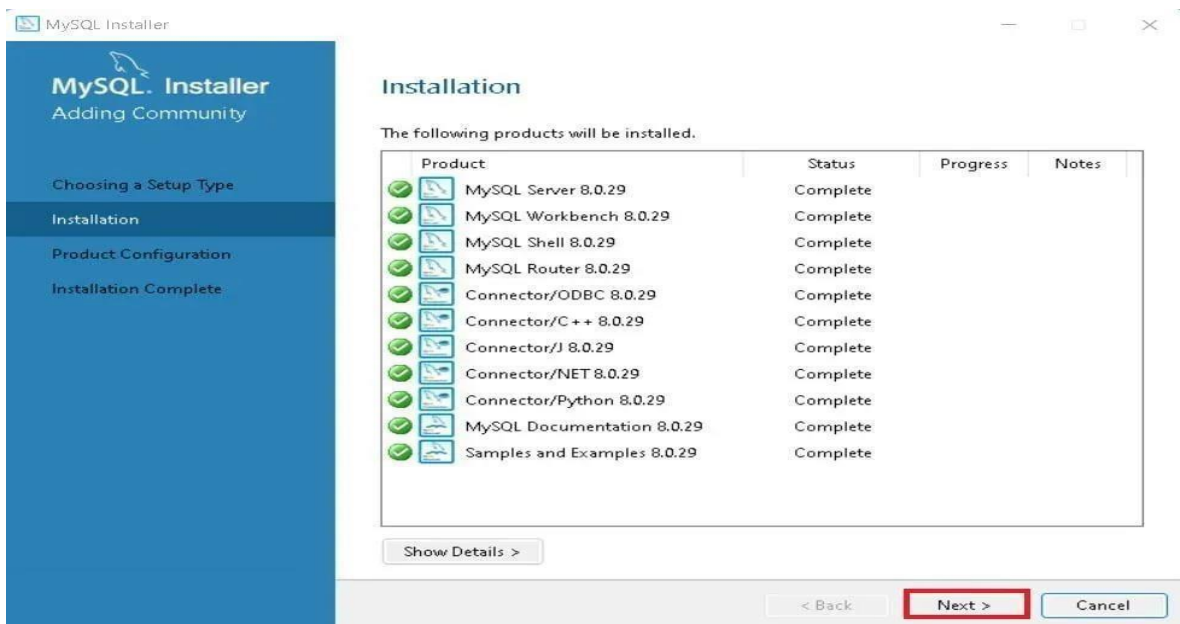
Step 6 – Download Packages

Click Execute to download and install the listed packages.



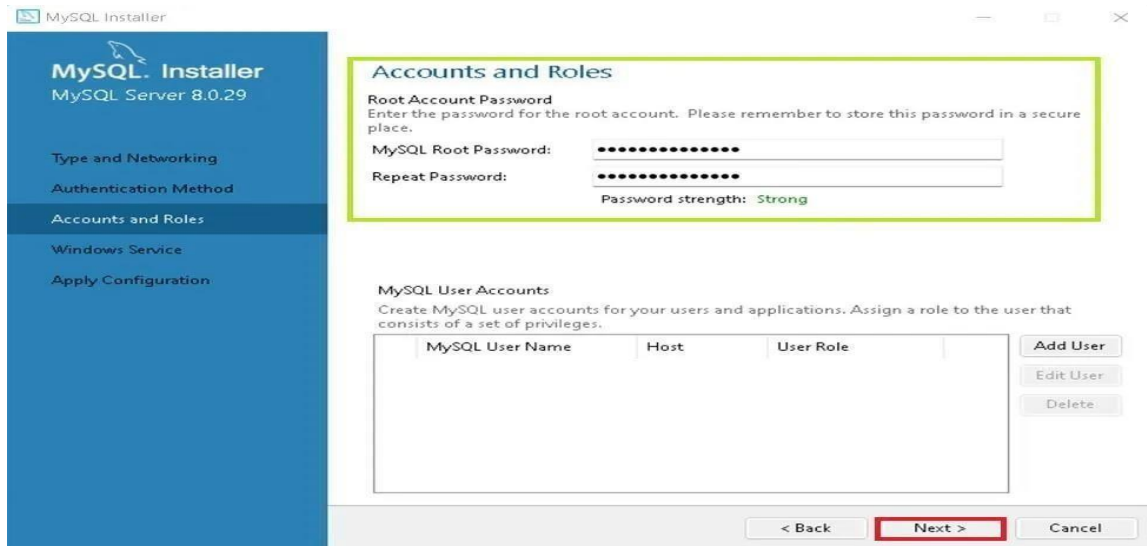
Step 7 – Installation

Wait until the listed products show Complete, then click Next.



Step 8 – Accounts and Roles

Enter the MySQL Root Password and repeat it.

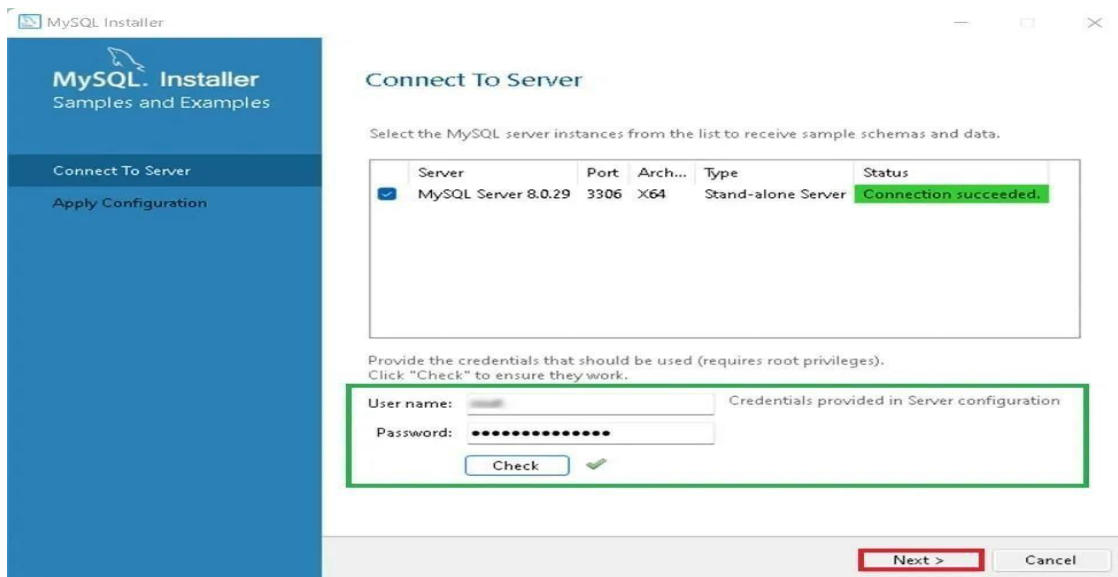


The screenshot shows the 'Accounts and Roles' step of the MySQL Installer. The left sidebar lists the installation steps: Type and Networking, Authentication Method, Accounts and Roles (selected), Windows Service, and Apply Configuration. The main area is titled 'Accounts and Roles' and contains a 'Root Account Password' section with two password input fields and a 'Password strength: Strong' indicator. Below this is a 'MySQL User Accounts' section with a table for adding users and buttons for 'Add User', 'Edit User', and 'Delete'. The 'Next >' button at the bottom is highlighted with a red box.

MySQL User Name	Host	User Role
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Step 9 – Connect to Server

Enter the credentials and click Check until the connection succeeds.



The screenshot shows the 'Connect To Server' step of the MySQL Installer. The left sidebar lists the installation steps: MySQL. Installer Samples and Examples, Connect To Server (selected), and Apply Configuration. The main area is titled 'Connect To Server' and contains a table for selecting server instances. The first instance, 'MySQL Server 8.0.29', is selected and has a status of 'Connection succeeded.'. Below the table is a 'User name' and 'Password' input section with a 'Check' button. The 'Next >' button at the bottom is highlighted with a red box.

Server	Port	Arch...	Type	Status
☒ MySQL Server 8.0.29	3306	X64	Stand-alone Server	Connection succeeded.

Step 10 – Installation Complete

Click Finish to complete the installation

